

Variables and Data types in Python

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$$s = a + b$$

↑
variable

A variable, which is represented by a letter or word, and it is used to represent unknown quantity or value.

$a = 10$
 $b = 15$
 $s = a + b$
↑
 $s = 10 + 15, \quad s = 25$

↑ Math
↑

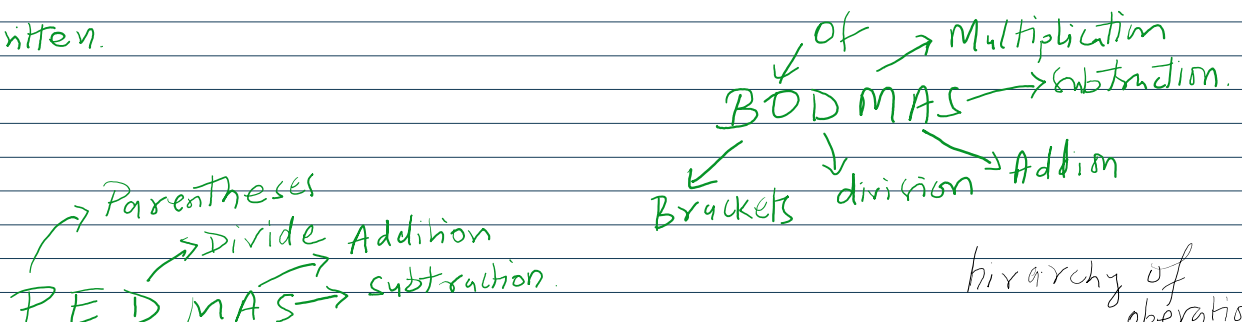
How to use this idea in computer?

In programming we use variables or objects to hold data or information in it. You can imagine, a variable like storage. We can say that it is a named location.



← These jars or storage units can be type of variable.

In variables, when new value comes old value is removed or overwritten.



PEDMAS

- Divide
- Addition
- Subtraction
- Multiplication
- Exponentiation

hierarchy of operation

This rule is called precedence of operator rule.

What is operator?

Operator is a symbol, which represents type of operation.

$S = a + b$

- operator
- operation (addition)
- operand or variable
- operator

(The job of this operator is called assignment)

It assigns the value from right to left.)

left = right

- always a variable
- a variable or an expression

$(a + b)$ } examples of
 $(a + b) * c$ } expression

```
>>> a = 10
>>> a
10
>>> b = 45
>>> b
45
>>> r = a * b
>>> r
450
>>>
```

Data type

What is data?

Ans:- Data is raw information.

What is information?

Ans:- It contains processed data, which have some meaning to it.

alphabets

name = 'vaanya' } data.

age = 11 } integer

price = '\$1000' } string

special symbol } numeric value

if these are stored inside a single quote, or double quote they are called strings

$a = 10$ } raw data
 $b = 5$ }
 $D = a * b$

which have some meaning to it.

$a = 10$ } raw data
 $b = 5$ }

$P = a \times b$
↑
processed data
information.

Python Data Types

Data types tell what kind of value a variable holds. Python is dynamically typed, meaning you don't need to declare types explicitly.

$$(N)^{1/2} \leftarrow \sqrt[2]{N} \leftarrow \text{square root.}$$
$$(N)^{1/3} \leftarrow \sqrt[3]{N} \leftarrow \text{cube root}$$

$$\sqrt{-25} = \sqrt{-1} \times \sqrt{25} = i \sqrt{5 \times 5} = 5i$$

$[N > 0]$
or a positive real number

opposite of real number
is imaginary number.

is called complex number

imaginary number
or
complex number.

In Python
 $5j$

Python is designed by Guido

'i' or 'I' is not good, it represents 'ego'
Guido removed 'i' or 'I'.

Instead of 'i', in Python Guido has introduced 'j'.

```

>>> r
450
>>> type(r)
<class 'int'>
>>> d = 3.46
>>> d
3.46
>>> type(d)
<class 'float'>
>>> f = 3j
>>> type(f)
<class 'complex'>
>>> g = 0j
>>> g
0j
>>> type(g)
<class 'complex'>
>>> h = 5 + 7j
>>> type(h)
<class 'complex'>
>>> h
(5+7j)
>>>

```

Numeric Types

1. int
2. float
3. complex

Text type or String type

```

>>> name = 'Vaanya'
>>> name
'Vaanya'
>>> sent = "Using Python programming language is fun"
>>> sent
'Using Python programming language is fun'
>>> sent1 = '''this is triple quote'''
>>> sent1
'this is triple quote'
>>> type(sent1)
<class 'str'>
>>> type(sent)
<class 'str'>
>>> type(name)
<class 'str'>
>>>

```

Boolean type.

The value will be either 'True' or 'False'.

```

>>> t = False
>>> t
False
>>> type(t)
<class 'bool'>
>>> t = True
>>> t
True
>>> type(t)
<class 'bool'>
>>> a
10
>>> b
45
>>> v = a > b
>>> v
False
>>> type(v)
<class 'bool'>
>>>

```

Basic data types in Python

Numeric type

```

├── int
├── float
└── complex

```

Text type

```

├── str

```

Boolean

```

├── bool

```

None

```

├── None

```

```

>>> product = None
>>> type(product)
<class 'NoneType'>

```